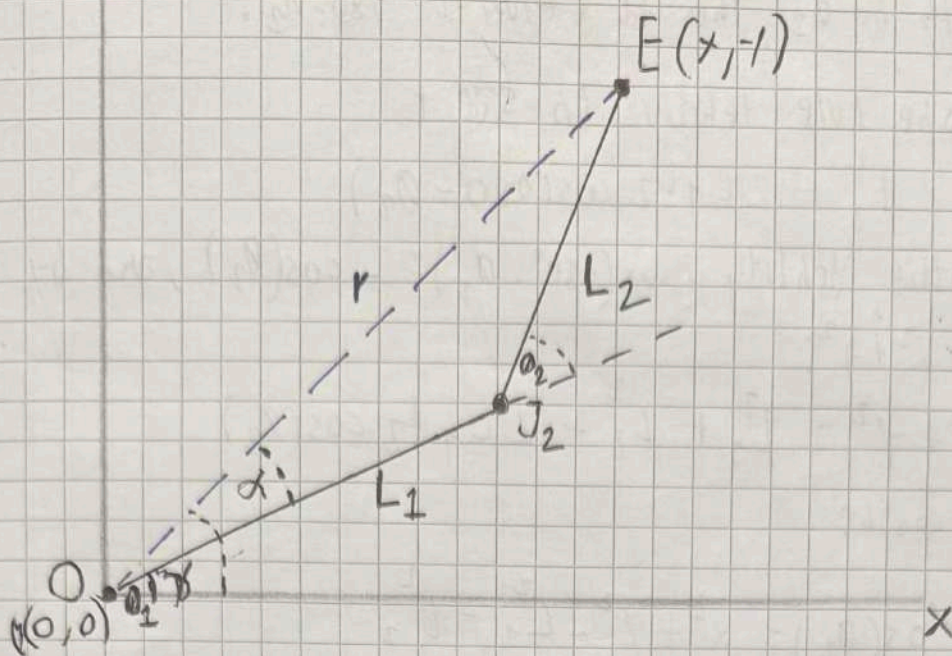




Let J_2 be the joint at the end of link 1

Let L_1 be the length of link 1

Let L_2 be the length of link 2

Let E be the end-effector target position at (x, y)

Let r be the distance from The Origin to E

Let O be the origin at $(0,0)$

Let θ_1 be the absolute angle of Link 1 relative to the positive x-axis

Let θ_2 be the relative angle of Link 2 measured relative to the extension of Link 1

Let α be the interior angle of triangle $\triangle O-J_2-E$ at the Origin opposite to L_2

Let γ be the total angle from the positive x-axis to the r

The target position vector r must satisfy the following domain constraint

$$|L_1 - L_2| \leq r \leq L_1 + L_2$$

If this condition is not met, then the inverse trigonometric function is invalid, and no geometric solution exist.

Using the Pythagorean theorem distance r is calculated as

$$r = \sqrt{x^2 + y^2}$$

Using the triangle $\Delta O-J_2-E$; the interior triangle angle at J_2 is supplementary to θ_2 , Thus it is equal to $180^\circ - \theta_2$.

Applying the cosine rule relative to side r

$$r^2 = L_1^2 + L_2^2 - 2L_1L_2 \cos(180^\circ - \theta_2)$$

Using trigonometric identity, $\cos(180^\circ - \theta_2) = -\cos(\theta_2)$, and by substituting $r^2 = x^2 + y^2$:

$$x^2 + y^2 = L_1^2 + L_2^2 + 2L_1L_2 \cos(\theta_2)$$

Then, isolating cosine

$$\cos(\theta_2) = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$$

Finally, by taking the inverse cosine

$$\theta_2 = \pm \cos^{-1} \left(\frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2} \right)$$

The angle γ is calculated using trigonometry as:

$$\gamma_r = \tan^{-1} \left(\frac{y}{x} \right)$$

Because $\tan^{-1}$ only returns value between -90° and 90° , it is unable to distinguish target in Quadrant II and III. Thus, a manual quadrant correction is used

X	Y	Quadrant	Corrected γ
+	+	I	$\gamma = \gamma_r$
-	+	II	$\gamma = \gamma_r + 180^\circ$
-	-	III	$\gamma = \gamma_r + 180^\circ$
+	-	IV	$\gamma = \gamma_r$

Applying the cosine rule for the interior angle α

$$L_2^2 = L_1^2 + r^2 - 2L_1r \cos(\alpha)$$

Isolating $\cos(\alpha)$ and substituting $r = \sqrt{x^2 + y^2}$ results in:

$$\alpha = \cos^{-1} \left(\frac{L_1^2 + x^2 + y^2 - L_2^2}{2L_1 \sqrt{x^2 + y^2}} \right)$$

θ_1 can be expressed as

$$\theta_1 = \gamma \pm \alpha$$

$$\theta_1 = \tan^{-1}\left(\frac{y}{x}\right) - \text{sgn}(\theta_2) \cos^{-1} \left(\frac{L_1^2 + x^2 + y^2 - L_2^2}{2L_1 \sqrt{x^2 + y^2}} \right)$$

Note: $\tan^{-1}\left(\frac{y}{x}\right)$ is subject to manual quadrant correction based on the sign of y and x . The signum function $\text{sgn}(\theta_2)$ subtract α for $\theta_2 > 0$ and add α for $\theta_2 < 0$.